

# Enhanced Techniques to Implement Jumping-Over-Down and Jumping-At-Air using Pressure-sensing Shoes in Virtual Reality

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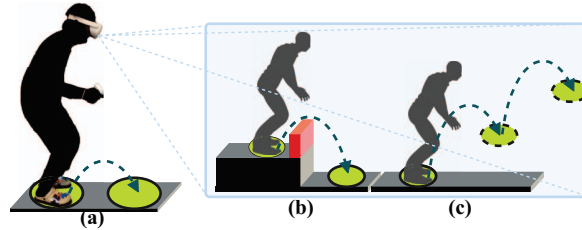


Figure 1: (a): A user wears virtual reality equipment and pressure-sensing shoes. The user jumps normally in the real world. In the virtual scene, the redirected jumping algorithm [2] drives the avatar. The avatar executes (b) Jumping-Over-Down for a downward movement or performs (c) Jumping-At-Air to hover and then jump again.

## ABSTRACT

Enhance the experience of jumping in the virtual reality environment is interesting and useful for various applications. This work includes three parts which are Pressure-sensing Shoes, Jumping-Over-Down and Jumping-at-Air. One particular designed pressure-sensing shoes are used for jumping detection. This design could substantially reduce the cost of detecting jumping behaviors while ensuring high precision. Jumping-Over-Down allows users to control a virtual avatar to perform realistic downward jump in virtual reality environment. While Jumping-at-Air, based on a novel redirection-jump mechanism, represents a type of surreal jumping which resembles the double-jump feature found in video games. From the hovering position, user can execute another jumping.

**Index Terms:** Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual Reality;

## 1 MOTIVATION

In virtual reality, Redirected Jumping [2, 5] adjusts users' jump trajectories to drive virtual avatars for more efficient and realistic vertical movement. However, existing jump detection algorithms lack accuracy and solely focus on reproducing physically plausible jumping behaviors in the virtual world. We propose a pressure-sensitive shoe-based algorithm that greatly enhances accuracy and introduces the Jumping-Over-Down jumping behavior. Additionally, we introduce a new surreal jumping method called Jumping-At-Air, expanding the boundaries of virtual reality jumping.

## 2 OVERVIEW

As illustrated in Fig. 1, the main components are divided into three parts:

(a) represents the jump detection algorithm based on pressure-sensing shoes. User wears a pair of customized shoes in the real world, and we determine if the user has jumped based on the pressure measured at the shoe's sole.

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(b) depicts an avatar performing a Jumping-Over-Down action in a virtual environment. Jumping-Over-Down refers to the avatar jumping over an obstacle and landing on a lower ground in front. The implementation of Jumping-Over-Down relies on a five-stage redirected jumping framework [2] and employs the jump detection algorithm from (a).

(c) shows the avatar executing the Jumping-At-Air action multiple times in a virtual environment. Jumping-At-Air means the avatar jumps, hovers mid-air, and can jump again while still in the air, similar to the double-jump mechanism seen in video games. Similar to [5], Jumping-At-Air also blends the *Ascending* and *Descending* stage of redirected jumping and introduces a customized redirection method. It employs the jump detection algorithm from (a), too.

## 3 PRESSURE-SENSING SHOES

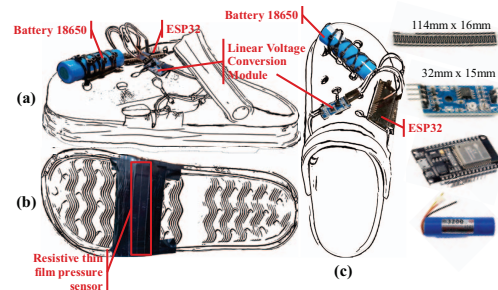


Figure 2: (a): The shoe equipped with a thin-film pressure sensor located at the sole, which efficiently captures pressure data. (b) and (c) show an ESP32 chip, powered by a battery, is responsible for gathering this pressure information. This chip also enables the seamless transmission of the data to a network via WiFi, ensuring timely and reliable data communication for further analysis and application.

In our study, we have developed a pressure-sensing shoe (Fig. 2) tailored for high-precision jump detection. The shoe equipped with a thin-film pressure sensor placed on the forefoot position of the shoe sole, which efficiently captures pressure data. An ESP32 [1, 3] chip, powered by a battery, is responsible for gathering this pressure information. This chip also enables the seamless transmission of

the data to a network via WiFi, ensuring timely and reliable data communication for further analysis and application.

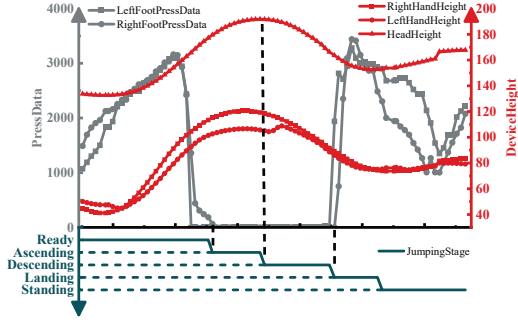


Figure 3: The precise timing of the transition in the jumping stage can be ascertained with accuracy through the utilization of pressure data and devices' height.

As show in Fig. 3, the moment the pressure sensor value swiftly decreases to 0, the jumping stage transitions from Ready to Ascending. Conversely, a rapid increase in the pressure sensor value signals the transition from Descending to Landing. The commencement of a decline in head height signifies the transition from Ascending to Descending.

#### 4 JUMPING-OVER-DOWN

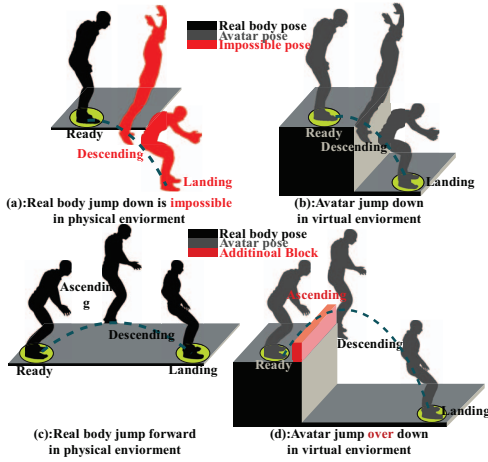


Figure 4: (a):Real body jump down is impossible in physical environment. (b):Jumping Down lacks the *Ascending* stage, it is not suitable for the redirected jumping algorithm. (c):User was guided to perform a normal jump. (d):Jumping-Over-Down has both *Ascending* and *Descending* Stage.

Jumping Down doesn't have the *Ascending* stage (Fig. 4 a), making the five-stage method unsuitable for virtual reality redirection. Additionally, users cannot perform Jumping Down through a flat ground (Fig. 4 b). This physical limitation poses a challenge in simulating certain aspects of jumping in virtual environments while maintaining a sense of realism.

Jumping-Over-Down incorporates an elevated obstacle strategically positioned at the edge (Fig. 4 c), prompting users to execute a conventional jumping comprising both *Ascending* and *Descending* stages in the real-world setting (Fig. 4 d).

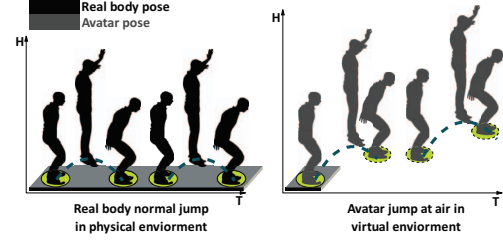


Figure 5: Incorporating the *Double Jumping* mechanism into the Virtual Reality Environment via Jumping Redirection.

#### 5 JUMPING-AT-AIR

The concept of Jumping-At-Air bears a striking resemblance to the *Double Jumping* mechanism found in gaming [4] : the avatar jumps and hovers mid-air, followed by a subsequent jump to reach higher elevations (Fig. 5).

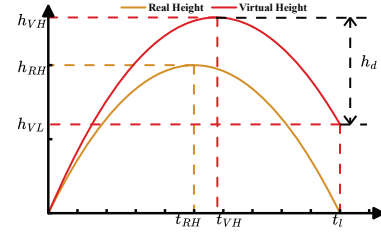


Figure 6: Simultaneous termination of virtual and actual jumps occurs; however, the virtual jump attains its zenith later, culminating in a higher final position.

The design of Jumping-At-Air is based on two fundamental principles (Fig. 6): (1). The height of the avatar's *Ascending* stage in the virtual world is directly correlated with the initial velocity of the real-world jumping; (2). The *Descending* distance of the avatar in the virtual world remains constant. Users can adjust the avatar's ascending height in the virtual world by adjusting the initial velocity of their real-world jumping. Given the fixed *Descending* distance, users can accurately predict the landing position.

$$h_v = 2kv_0t - \sqrt{2kg_r h_d}t - \frac{1}{2}kg_r t^2 \quad (1)$$

We ultimately employ formula Equation 1 to drive the virtual avatar for Jump-At-Air.

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